

Flexible Coherent PON for 6G Fixed Networks: Technologies and Solutions

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ABSTRACT

The forthcoming sixth-generation fixed (F6G) network promises substantial advancements in capacity, connectivity, and latency. Owing to the high receiver sensitivity and spectral efficiency, flexible coherent passive optical networks (PON) offer the potential to significantly increase access performance, delivering enhanced capacity, expanded connectivity, and reduced latency for the evolution to the F6G network. The flexible coherent PON will apply several emergent technologies, including coherent optics technology, flexible multiple access, and flexible-rate technology. As an important application scenario of the flexible coherent PON, coherent point-to-multi-point (P2MP) optical aggregation can potentially replace point-to-point electrical aggregation for the F6G network, which enables the flexible upgrade and has low operational and capital expenditure. Based on the coherent P2MP optical aggregation, we demonstrate flexible multiple access of joint frequency-division multiple access and time-division multiple access (TDMA) cooperating with flexible-rate technology to balance capacity, connectivity, and latency. Meanwhile, burst-mode digital signal processing is proposed for the TDMA burst signal in the flexible multiple access. In summary, the flexible coherent PON is poised to make significant contributions to the future F6G network.

INTRODUCTION

Fixed networks play an essential role in a new communication era by providing high bandwidth and availability for satisfying future bandwidth-hungry applications, including ultra-high-definition video streaming, telemedicine, and virtual reality gaming. Therefore, the evolution of the fixed network is always underway[1]. In 2020, the European Telecommunication Standards Institute (ETSI) officially established the F5G industry working group, which is dedicated to promoting the progress from fiber-to-the-home to fiber-to-the-everything, including fiber-to-the-office, fiber-to-the-room, fiber-to-the-desk, and fiber-to-the-machine [2]. F5G network has the main features of full-fiber connection, enhanced fixed broadband, and guaranteed reliable experience, which aims to maximize the synergy with the optical transport network (OTN),

optical access network, and wireless local areas network such as 200G/400G optical transport network, XG(S)-passive optical network (PON), and Wireless Fidelity (WiFi) 6 [3]. In 2024, ETSI finalized the release of F5G advanced network specification, which includes 400G/800G OTN, 50G PON, and WiFi 7. In the future sixth-generation fixed (F6G) network, it can be forecasted that the capacity, connectivity, and latency improvement will be more than 10 times [4].

To achieve the visions of the F6G network, Beyond 50G-PON plays a critical role. The direct-detection (DD) PON used in the previous-generation fixed networks is challenging to achieve the features of the F6G network due to the limited receiver sensitivity and spectral efficiency. Owing to the high receiver sensitivity and spectral efficiency, flexible coherent PON will be explored in the Beyond 50G-PON, including more innovative technologies such as coherent optics technology, flexible multiple access, and flexible-rate technology, making the capacity, connectivity, and latency possible to reach the targets for the F6G network. It is worth noting that the flexible coherent PON still faces many challenges before its practical applications, including how to balance the capacity, connectivity, and latency and implement burst-mode digital signal processing (DSP) for flexible multiple access.

In this article, we first introduce the potential technologies, including coherent optics technology, flexible multiple access, and flexible-rate technology for flexible coherent PON. As an important application scenario of the flexible coherent PON, coherent point-to-multi-point (P2MP) optical aggregation can potentially replace the point-to-point (P2P) electrical aggregation for the F6G network. Based on the coherent P2MP optical aggregation, we demonstrate joint frequency-division multiple access (FDMA) and time-division multiple access (TDMA) cooperating with flexible-rate technology, which can balance capacity, connectivity, and latency. Finally, we propose a burst mode DSP to deal with TDMA burst signal in the flexible multiple access.

KEY TECHNOLOGIES FOR FLEXIBLE COHERENT PON

Figure 1 depicts the evolution and trend of the ITU-T standards, indexes, and technologies for

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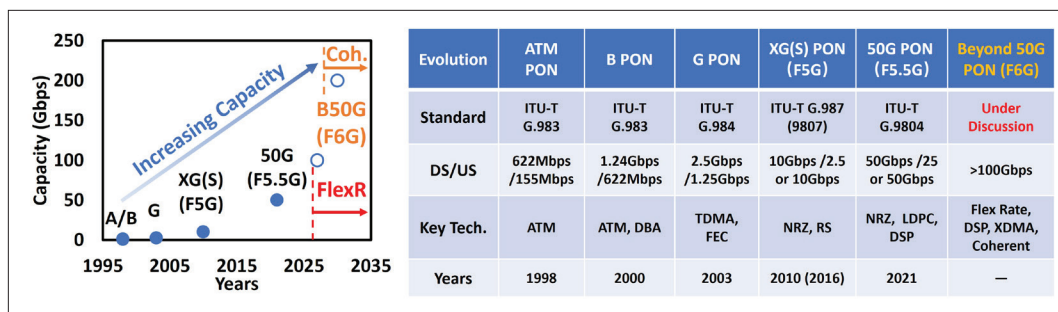


FIGURE 1. Evolution and trend of the ITU-T standards, indexes, and technologies for PON, which meet the requirements of the capacity-increasing and more flexible broadband access services.

PON, which meet the requirements of the capacity-increasing and more flexible broadband access services [5]. The ITU-T has standardized several generations of PON in the past three decades. In 1998, the first commercial PON was standardized as the asynchronous-transfer-mode PON (ATM-PON) by ITU-T G.983. ATM-PON is a part of broadband PON (B-PON), which has maximum downstream and upstream data rates of 622Mb/s and 155Mb/s, respectively. ATM-PON defines the data, voice and video control functions, security features, dynamic bandwidth allocation (DBA), and so on. B-PON inherited the ATM frame and further developed it in 2000, with maximum downstream and upstream data rates of 1.24Gb/s and 622Mb/s, respectively. Unlike the ATM-PON, B-PON added an extra wavelength for video. In 2003, the Gigabit PON (G-PON) was standardized by ITU-T G.984, the widest deployed PON. The maximum downstream and upstream data rates of G-PON are 2.5Gb/s and 1.25Gb/s, respectively. In G-PON, the uplink uses the TDMA, which requires burst-mode optical and electrical technologies. Meanwhile, with the increase in data rate, the forward error correction (FEC) technology was applied to ensure the performance. In 2010, the XG-PON system was standardized by ITU-T G.987. The maximum downstream and upstream data rates of XG-PON are 10Gb/s and 2.5Gb/s. In 2016, the XG(S)-PON system was standardized by ITU-T G.9807. The maximum downstream and upstream data rates of XG(S)-PON are both 10Gb/s. In 2019, hybrid time-and-wavelength division multiplexing (TWDM) PON was standardized by ITU-T G.989.2, which can achieve a nominal aggregate capacity of 40Gb/s (i.e., $4 \times 10\text{Gb/s}$). In recent years, XG(S)-PON has been the largest-scale product supplied to network operators. Based on XG(S)-PON, ETSI has started to promote the standard of the F5G network. In 2021, the 50Gigabit PON (50G-PON) system was standardized by ITU-T G.9804. Beginning in 2023, the commercial products of 50G-PON were gradually completed and practically applied. The DSP is first applied in the 50G-PON to improve performance. Based on 50G-PON, ETSI pushes forward the F5G advanced network. Furthermore, the ITU-T has established a study group to discuss Beyond 50G-PON. At the same time, the ETSI announces the visions of the F6G network. A leapfrog development will appear on Beyond 50G-PON to meet the visions of the F6G network.

Depending on the evolutionary law of the ITU-T standards, the Beyond 50G-PON may increase the data rate four times than 50G-PON. Therefore, the Beyond 50G-PON will achieve a data

rate of 200Gb/s. There are two technological paths for the Beyond 50G-PON: DD optics technology and coherent optics technology. Both previous PONs employ DD optics technology. In 50G PON, the optical power budget is still tight to meet the requirement of at least 32 dB, although the DSP and high-power laser are adopted. It can be expected that the optical power budget is more challenging to meet the requirement of 32dB if the DD optics technology is still employed in Beyond 50G-PON unless a laser with more increased power is adopted at the transmitter and optical amplifier is used at the receiver [6]. Due to the contradiction between the optical power budget and spectral efficiency, the burst devices' bandwidth should be further increased to meet the requirement of 200G burst signal reception. Meanwhile, the DD PON uses the TDMA or TDMA with wavelength division multiplexing, which faces the challenge of high latency or additional replacement cost and link loss, respectively.

Owing to the tough challenges of the DD optics technology, the coherent optics technology will potentially be adopted in Beyond 50G-PON. The coherent receiver has a naturally high sensitivity to increase the optical power budget and can recover high spectral-efficiency optical signals with polarization multiplexing and in-phase/quadrature modulation to relieve the stress of devices. Meanwhile, coherent optics technology makes it feasible to realize flexible multiple access, such as FDMA, and its integration with TDMA to balance the connectivity and latency for the Beyond 50G-PON. Moreover, the flexible-rate technology can be used to deliver flexible-rate data that matches the optical power budget of each optical network unit (ONU), which further increases the overall capacity. Figure 2 shows the key technologies for future flexible coherent PON, including coherent optics technology, flexible multiple access, and flexible-rate technology.

COHERENT OPTICS TECHNOLOGY

Around 2020, the 800Gb/s coherent optics technology was realized for P2P and P2MP optical communications, which met the rapid growth of network capacity and made the optical network more flexible[7]. Shortly, 1.6Tb/s coherent optics technology will be developed to meet the requirement for fast-growing data transmission. The advanced optoelectronic integrated circuit and higher-integration DSP ASIC technologies significantly reduce coherent optical transceivers' cost, footprint, and power consumption. Therefore, the applications of coherent optics technology are

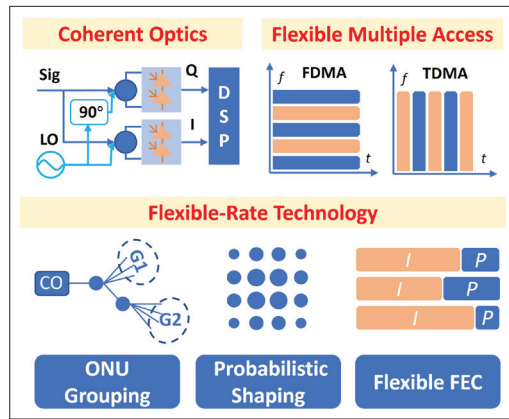


FIGURE 2. The key technologies for the flexible coherent PON including coherent optics technology, flexible multiple access, and flexible-rate technology.

extended to data-center networks and metro networks. Furthermore, the coherent optics technology has the potential to be applied in the future Beyond 50G-PON [8]. However, the coherent transceiver has a relatively high cost, which is unsuitable for the cost-sensitive ONU. Moreover, the coherent DSP has a high power consumption and a long convergence time. In conclusion, the coherent PON should establish competitive advantages before its commercial applications, which require a low-cost coherent transceiver and fast-convergence and low-complexity DSP. The simplified coherent transceiver and matched DSP architecture have been introduced in detail in our previous work [9]. In the following section, we will introduce the burst-mode coherent DSP to decrease the convergence time and evaluate its performance.

FLEXIBLE MULTIPLE ACCESS

TDMA was practically applied to optical access networks. Owing to the statistical multiplexing, the DBA of TDMA makes overall throughput higher. For the downlink, ONU should use high-bandwidth devices to receive the signals of all the ONUs. However, only the signal of one ONU is extracted from its time slot; thus, the rest of the time slots are discarded. For the uplink, high-bandwidth burst-mode devices are required to receive the burst signals from all the ONUs. Many technical problems have yet to be solved for the high-bandwidth burst-mode devices. Meanwhile, the TDMA makes the latency high for the ONUs, which is unsuited for latency-sensitive services.

FDMA and code-division multiple access (CDMA) can avoid the mentioned problems in TDMA, which are widely applied in commercial 1G-5G wireless access networks. The FDMA and CDMA have been widely studied but have not been used in the commercial DD PON. In the commercial DD PON, the ONUs use low-cost lasers with unstable wavelengths to reduce cost. When the FDMA and CDMA are adopted, the upstream optical carriers of the ONUs beat at the optical line terminal (OLT). The beat signals fall on the useful in-band signals to seriously degrade the performance. This is one of the reasons why the TDMA is the preferred option for the commercial DD PON. In the TDMA, the laser of one ONU works, and the lasers of the other ONUs are turned off to avoid the beat signal.

In the coherent PON, coherent optics technology can avoid the beat signal between the optical carriers, making it possible for the FDMA and CDMA practical applications. For implementing flexible multiple access, the FDMA, CDMA, and TDMA can be integrated to balance connectivity and latency. Flexible multiple access can make the future PON more flexible than the existing PON. Compared to the TDMA, CDMA cannot reduce the device bandwidth of the ONUs due to the spectrum spreading. FDMA can use the devices with bandwidth matching the bandwidth of the subcarrier, which partly reduces the cost of devices for the cost-sensitive ONU [9]. In the following section, we will introduce flexible multiple access based on joint FDMA and TDMA cooperating with flexible-rate technology to balance the capacity, connectivity, and latency.

FLEXIBLE-RATE TECHNOLOGY

In the deployed PON, the optical signals from OLT to ONUs pass the different lengths of fiber and different numbers of optical splitters. Therefore, the link losses between the OLT and ONUs are different. The statistics of the commercially deployed ONUs show that the percentage is less than 3 percent for the ONUs with an ROP less than -24dBm , and the ROP of more than 80 percent of ONUs is larger than -22dBm [10]. The data rate of the deployed ONUs is limited by the optical power budget of the worst-case ONU, for each ONU works at the same pre-defined data rate, which implies insufficient exploitation of the optical power budget. To overcome the existing drawbacks, flexible-rate technology has been proposed to deliver flexible-rate data to match the optical power budget of each ONU, which has the potential to increase the overall capacity for the future PON. In the flexible-rate PONs, the time frame or bandwidth is subdivided into time slots or frequency subcarriers for mapping the ONUs. The signals with different spectral efficiencies are assigned to the different time slots and frequency subcarriers with different data rates.

ONU Grouping: In theory, the data-rate adjustment for flexible-rate PONs should be continuous with fine granularity to meet the link losses of all ONUs. The probabilistic constellation shaping pulse amplitude modulation (PCS-PAM), discrete multi-tone (DMT), and FEC are candidate schemes for the flexible-rate PONs. The distribution matcher, such as the constant composition distribution matcher (CCDM), adjusts the spectral efficiency of PCS-PAM [11]. The DMT can change spectral efficiency based on the bit loading for the fine-granularity data-rate adjustment [12]. The shortening and puncturing operation for FEC can change the net data rate of the coded PAM [13]. However, taking into account the complexity and feasibility, the levels of data-rate adjustment are enumerable for practical scenarios. The ONUs connecting the same optical splitter have almost the same link losses. To match the enumerable levels of data-rate adjustment, the ONUs in the flexible-rate PONs should be grouped depending on their link losses. The ONUs in one grouping can allocate signals with the same spectral efficiency which have nearly the same optical power budget. The idea of the ONU grouping is similar to cell allocation in wireless access networks.

PCS Modulation: PCS modulation has a natural feature for the flexible adjustment of spectral efficiency. PCS modulation has been widely applied to improve the performance of long-haul coherent optical communication. Due to the use of an Erbium-doped fiber amplifier (EDFA), long-haul coherent optical communication can be considered an average-power-constrained (APC) system. In the APC system, the PCS modulation has a larger Euclidean distance than the same-order square modulation to obtain a better performance. However, in the PON scenario, the EDFA is unpopular due to its high cost. Therefore, the PON without the EDFA can be classified as a peak-power constrained (PPC) system. In the PPC PON, single-carrier and multi-carrier modulations should be analyzed separately.

As the single-carrier modulations, the inverse PCS-PAM performs better than the common PCS-PAM for the PPC system [11]. Therefore, the inverse PCS-PAM has been used to adjust the spectral efficiency for flexible-rate adjustments [13]. As the multi-carrier modulations, the PCS-quadrature amplitude modulation (PCS-QAM) is modulated on the frequency-domain subcarriers as the entropy-loading operation. In the entropy-loading multi-carrier modulations, the PCS-QAM should be normalized to achieve the same unit average power, which can be considered as an APC system. Therefore, the common PCS-QAM has been applied in the entropy-loading multi-carrier modulations to adjust the spectral efficiency [12]. In conclusion, the PCS-based single-carrier and multi-carrier modulations can flexibly adjust the spectral efficiency for realizing the flexible-rate PONs.

Flexible FEC Technology: The PCS modulation and flexible FEC technology can be combined to achieve more linear adjustment [13]. With the data rate increase to 50Gb/s, high-performance soft-decision FEC with 20 percent overhead has been standardized for 50G PON to increase its optical power budget. In general, the pre-FEC BER of hard-decision FEC with 7 percent overhead is 3.8×10^{-3} , and the pre-FEC BER of soft-decision FEC with 20 percent overhead is 2×10^{-2} . The optical power budget can be improved by more than 2dB when the pre-FEC BER is changed from 3.8×10^{-3} to 2×10^{-2} . The FEC overhead can be used to exchange the optical power budget to meet the requirements of the flexible-rate PON. The simplest method for achieving different FEC overheads is to use the FECs with different code words. However, using the different code words in one PON is difficult in the practical scenario. Compared to the change of code word, the shortening and puncturing operations are more reasonable for the flexible FEC. When only one code word is used, shortening and puncturing operations can continuously change the payload and overhead of the FEC to achieve different code rates. With the decrease in the code rate, the optical power budget increases, and the net data rate decreases. Based on the flexible FEC technology, the net data rate can be adjusted from 88.89Gb/s to 73.33Gb/s for 100Gb/s PAM4 and from 44.44Gb/s to 36.67Gb/s for 50G OOK [13].

COHERENT P2MP OPTICAL AGGREGATION

As an important application scenario of the flexible coherent PON, coherent P2MP optical aggregation

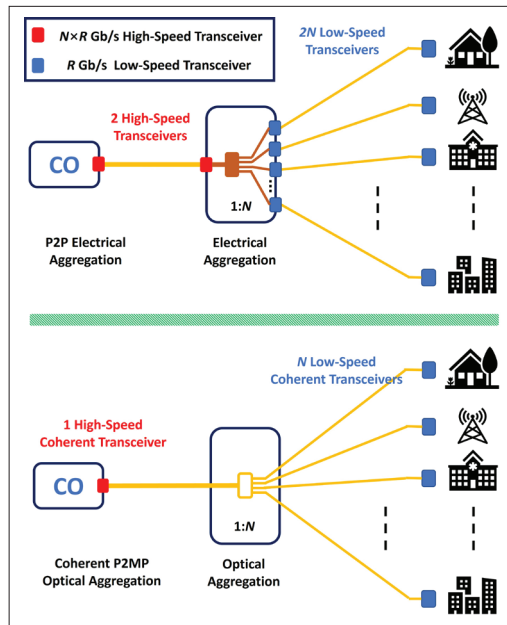


FIGURE 3. The P2P electrical aggregation and coherent P2MP optical aggregation for the F6G network.

gation can potentially replace the P2P electrical aggregation for the F6G network. Figure 3 shows the P2P electrical aggregation and coherent P2MP optical aggregation for the F6G network. In the P2P electrical aggregation, $2N$ of R Gb/s low-speed transceivers connect the leaves with the aggregation equipment. The electrical aggregation equipment, such as the router, converges the N of R Gb/s electrical signal flows into one $N \times R$ Gb/s electrical signal flow. Then, two $N \times R$ Gb/s high-speed transceivers connect the aggregation equipment with the central office (CO). The capacity of the high-speed transceiver must be higher than or equal to the aggregate capacity of $2N$ low-speed transceivers. In the coherent P2MP optical aggregation, the optical aggregation based on a passive optical splitter converges N of R Gb/s optical signal flows into one $N \times R$ Gb/s optical signal flow. N low-speed coherent transceivers and one high-speed coherent transceiver connect the leaves with the CO through the passive optical splitter.

The advantages of the coherent P2MP optical aggregation are as follows [14]:

- The coherent transceivers are required at the leaves of the coherent P2MP optical aggregation. The coherent transceivers achieve a high capacity and optical power budget but are expensive. Fortunately, the number of high-speed and low-speed transceivers is 50 percent less than that in the P2P electrical aggregation, which can effectively reduce operational expenditure (OPEX) and capital expenditure (CAPEX).
- The optical aggregation uses only a passive optical splitter, which is much cheaper and more power-efficient than electrical aggregation using a router. Meanwhile, the transceivers at the leaves directly connect to the transceiver at the CO in the coherent P2MP optical aggregation, which naturally has a much lower latency than the P2P electrical aggregation using a router.

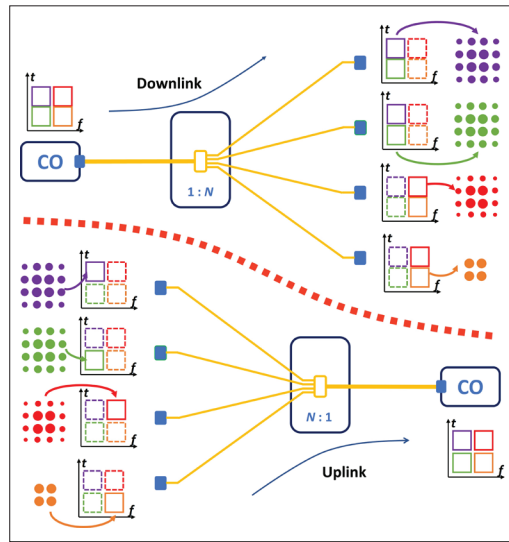


FIGURE 4. The P2MP (downlink) and MP2P (uplink) architectures of joint FDMA and TDMA integrating with flexible-rate technology.

- To increase the network capacity of the P2P electrical aggregation, the low-speed transceivers, high-speed transceivers, and router equipment should be upgraded simultaneously. In the coherent P2MP optical aggregation, the network capacity can be flexibly upgraded by adding the digital subcarriers when the different generations have the same definition of digital subcarriers. The capacity expansion can be implemented by one or more operations, such as using a higher-ratio passive optical splitter, adding more subcarriers at the leaves, and expanding transceiver capacity at the CO. In conclusion, the network capacity can be smoothly and flexibly upgraded in the coherent P2MP optical aggregation.

The challenges of coherent P2MP optical aggregation are as follows [14]:

- The P2P electrical aggregation leveraging a router offers a greater degree of flexibility, enabling a richer feature set, including functionalities such as authentication. Conversely, coherent P2MP optical aggregation based on passive optical splitters is inherently inflexible and lacks programmability for operators. Furthermore, P2P electrical aggregation exhibits superior reliability compared to its optical counterpart. This stems partly from the difficulty of implementing effective failure monitoring in coherent P2MP optical aggregation.
- The P2P electrical aggregation can regenerate the electrical signal to ensure the quality of the forwarded signal. However, when coherent P2MP optical aggregation is employed, the optical signal is directly transmitted between the CO and leaves. Therefore, the optical signal in the coherent P2MP optical aggregation suffers from more link losses, including fiber loss and optical splitter loss. The extra optical amplifiers may be employed in the coherent P2MP optical aggregation to ensure performance.
- The P2P electrical aggregation can provide fine granularity and multiple-user data aggregation. Currently, the coherent P2MP

optical aggregation utilizes the FDMA integrating with flexible-rate technology. When the subscriber number increases, the subcarrier number increases, but the subcarrier bandwidth decreases. However, the performance of low-bandwidth subcarriers is highly susceptible to the phase noise of the laser. Therefore, the subscriber number of the existing coherent P2MP optical aggregation based on FDMA is limited.

JOINT FDMA AND TDMA FOR FLEXIBLE MULTIPLE ACCESS

The TDMA can allocate the time slots on one frequency subcarrier to ensure sufficient connections for the subscribers. TDMA and FDMA are integrated to provide sufficient connections and low latency. For latency-sensitive subscribers, the FDMA provides independent subcarriers. Meanwhile, the TDMA can increase the subscriber number and bandwidth utility efficiency owing to the statistical multiplexing.

Figure 4 shows the P2MP (downlink) and MP2P (uplink) architectures of joint FDMA and TDMA integrating with flexible-rate technology based on entropy-loading digital subcarrier multiplexing (DSCM). The P2MP (downlink) architecture broadcasts the optical DSCM signal from the CO to the ONUs. At the ONU, the local oscillator aligns with the center wavelength of the optical subcarrier, which can downconvert the optical signal to the baseband electrical signal by coherent detection. The center wavelength of the optical subcarrier equals the center wavelength of the optical DSCM signal plus the wavelength shift (i.e., frequency shift of the subcarrier). The ONU only receives its corresponding subcarrier, which uses the transceiver with the bandwidth matching the subcarrier width. Therefore, the transceiver bandwidth at ONUs of the FDMA is relatively lower than that at ONUs of the TDMA. Integrating with the TDMA, the subcarrier can be divided into some time slots to provide services for more users.

In the MP2P (uplink) architecture, the optical signals are transmitted from ONUs to the CO. The subcarriers and time slots are allocated to ONUs. Then, the PCS-QAM signal is assigned to the subcarrier and time slot. The transceiver at the leaf generates one optical subcarrier, which only needs the bandwidth matching the subcarrier width. The minimum wavelength gap between the optical subcarriers should be larger than the signal bandwidth. The optical signals of the ONUs are aggregated to generate an optical DSCM signal by the optical splitter. At the CO, a transceiver for optical DSCM signal recovers the signals from all the ONUs. In conclusion, the transceiver bandwidth for an optical DSCM signal should match the bandwidth of whole subcarriers. By contrast, the transceiver at the leaf has lower bandwidth and cost.

For both P2MP (downlink) and MP2P (uplink), the flexible-rate technology based on entropy-loading DSCM can assign best-matched spectral efficiencies to the users based on their optical link losses, which can further increase the network capacity. In conclusion, the flexible multiple access of joint FDMA and TDMA integrating with flexible-rate technology can balance capacity, connectivity, and latency for the flexible coherent PON.

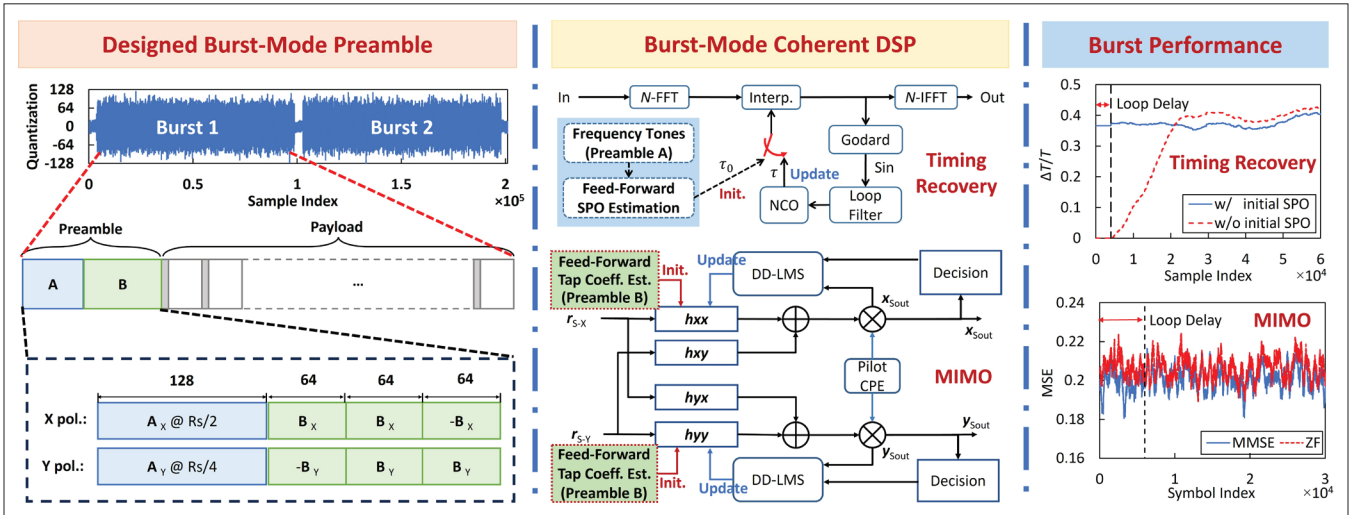


FIGURE 5. The burst preamble design, burst-mode timing recovery, and the burst-mode multiple-input-multiple-output (MIMO) equalizer for upstream reception in the coherent PON. By using the designed burst preamble and feed-forward estimation, the sampling phase offset (SPO) and tap coefficients are estimated promptly to eliminate the convergence time for the burst-mode timing recovery and burst-mode MIMO equalizer, respectively.

BURST-MODE DSP FOR COHERENT TDMA

The continuous-mode DSP can process the broadcast signal in the coherent PON, similar to that in long-haul optical transmission. However, coherent TDMA still requires a burst-mode DSP to process the upstream burst signal. Therefore, this section will introduce the burst-mode DSP for the coherent TDMA in detail.

In the continuous-mode DSP, the timing recovery and multiple-input-multiple-output (MIMO) equalizer use the blind algorithms with feedback updating loops to converge the sampling phase offset (SPO) and tap coefficients, respectively. The loop delays lead to extended convergence times for both the timing recovery and MIMO equalizer. The time duration of one burst upstream frame is fixed in the PON scenario. The longer convergence time means a shorter effective time for the payload, leading to a lower spectral efficiency. Therefore, the continuous-mode DSP is unsuitable for processing the upstream burst signal. The burst-mode DSP should be investigated to reduce the convergence time for the coherent PON. Particularly, the timing recovery and MIMO equalizer in the burst-mode DSP are the kernel modules that determine the convergence time. The burst-mode timing recovery and burst-mode MIMO equalizer use the designed burst-mode preamble and feed-forward estimation to avoid the feedback updating loops and shorten the convergence time.

Figure 5 shows the burst-mode preamble, burst-mode timing recovery, and the burst-mode MIMO equalizer for upstream reception in the coherent PON [15]. The designed burst-mode preamble consists of Preamble A and Preamble B for burst-mode timing recovery and burst-mode MIMO equalizer, respectively. In Preamble A with 128 symbols, the frequency tone A_X at half of the baud rate is designed for the X polarization. The frequency tone A_Y at a quarter of the baud rate is designed for Y polarization. Preamble B consists of three sets of 64-symbol constant-amplitude-zero-autocorrelation (CAZAC) sequences $B_{X,Y}$. For the X polarization, the three sets of 64-symbol CAZAC sequences B_X are multiplied by the coefficients of 1, 1, and -1. For the Y polarization, the three sets of 64-symbol CAZAC sequences B_Y are multiplied by the coefficients of -1, 1, and 1. The multiplication coefficients can improve the performance of frame synchronization and feed-forward tap coefficient estimation.

Based on Preamble A, frame detection, frequency offset estimation (FOE), one-tap state-of-polarization (SOP) estimation, and feed-forward SPO estimation can be implemented. The feed-forward SPO estimation has almost no convergence time based on the Preamble A. The estimated SPO is fed into the interpolator to compensate the SPO. Then, a feedback loop is used to update the SPO, which consists of the Godard algorithm, loop filter, and numerically controlled oscillator (NCO). The loop delay was set to 2000 symbols. As the experimental results show, the SPO updating based on the feedback loop does not work initially due to the loop delay. When the SPO is initialized correctly, there is no convergence time for the feedback loop. However, when the SPO is not initialized, the feedback loop requires a convergence time of 20000 symbols. The more accurate initialized SPO leads to a better performance at the beginning of the burst frame. Therefore, the timing recovery with SPO initialization can converge faster than that without SPO initialization. In conclusion, the feed-forward SPO estimation based on Preamble A is crucial for the burst-mode timing recovery.

Based on Preamble B, frame synchronization and feed-forward tap coefficient estimation can be realized. The feed-forward tap coefficient estimation can be implemented by minimum-mean square error (MMSE) or zero-forcing (ZF) algorithms. The feed-forward tap coefficient estimation has almost no convergence time based on Preamble B. The estimated tap coefficients are used to initialize the MIMO equalizer. Then, the tap coefficients of the MIMO equalizer are updated by a feedback loop based on the decision-directed least mean square (DD-LMS) algorithm with pilot-based carrier-phase estimation (CPE). The mean square error (MSE) curve of the MIMO equalizer can be used to evaluate the accuracy of initialized tap coefficients estimated by MMSE or

The flexible coherent PON is highly anticipated for the F6G network owing to innovative technologies such as coherent optics technology, flexible multiple access, and flexible-rate technology. As one of the application scenarios, the flexible coherent PON, coherent P2MP optical aggregation has the potential to replace P2P electrical aggregation, which can cut down OPEX and CAPEX.

ZF algorithms using Preamble B. The first symbols within the loop delay use the initialized tap coefficients. The MSEs of the first symbols within the loop delay have the same level compared to the MSEs of the subsequent symbols using the updated tap coefficients. It indicates no convergence time for the feedback loop when the tap coefficients are initialized correctly. It is worth noting that the ZF algorithm can be realized using Preamble B under a low computational complexity owing to the SOP estimation and recovery using Preamble A. However, the MSE of the ZF algorithm is still higher than that of the MMSE algorithm. In conclusion, if the performance is given priority, the MMSE algorithm is better than the ZF algorithm. When the computational complexity is preferred, the ZF algorithm can be chosen.

CONCLUSION

It is forecasted that the capacity, connectivity, and latency will be improved more than 10 times for the F6G network. The flexible coherent PON is highly anticipated for the F6G network owing to innovative technologies such as coherent optics technology, flexible multiple access, and flexible-rate technology. As one of the application scenarios, the flexible coherent PON, coherent P2MP optical aggregation has the potential to replace P2P electrical aggregation, which can cut down OPEX and CAPEX. Based on the coherent P2MP optical aggregation, flexible multiple access of joint FDMA and TDMA cooperating with flexible-rate technology is proposed to balance capacity, connectivity, and latency. The burst-mode DSP has been verified to fasten the convergence for processing the TDMA signal in flexible multiple access. In conclusion, the flexible coherent PON will bring more noteworthy additions to the future F6G network.

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